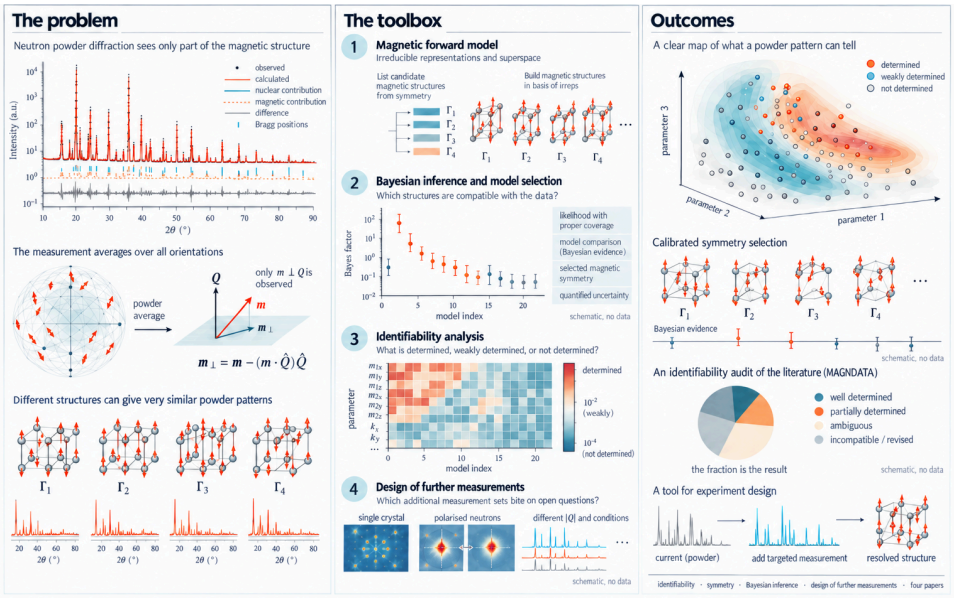
What a powder pattern can tell

Identifiability, Bayesian inference and symmetry in the determination of magnetic structures

ABSTRACT

More than two thousand magnetic structures are published, most from neutron powder diffraction, and the database collecting them reports that many were described ambiguously; even NiO and MnO were revised in 2024. The measurement is the reason: the neutron sees only the component of the moment perpendicular to the scattering vector, and the powder average discards orientation, so part of a structure is not in the data, part only weakly, and part of what is reported is the model's choice. Symmetry tools list the candidates; the choice is still made by agreement factors and judgement. This thesis asks which features of a magnetic structure a powder pattern determines, which it does not, and what cuts the rest, for one pattern and for the literature. Four contributions: a magnetic forward model in irreducible-representation coordinates, in superspace, that prints the directions the data do and do not fix; magnetic symmetry chosen by Bayesian evidence with a likelihood whose coverage is tested; an identifiability audit of the commensurate structures of MAGNDATA; and which further measurement settles what the powder leaves open. Expected outcomes: a diagnostic in an open engine, calibrated symmetry selection on the group's data, the identifiability map of the literature, a design tool for beam time, four papers.



Graphical abstract. Every plot is schematic and says so; the fraction in the audit is the result, not a value assumed.

Related tools. diff-ref-struct: jmigartua.github.io/diff-ref-struct-app · Ispilu: jmigartua.github.io/ispilu

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Chemistry, Materials Engineering or a related field, and a Master's degree in Physics, Materials Science, Nanotechnology or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in solid-state physics, magnetism, crystallography or diffraction; having used a refinement program (FullProf, GSAS-II or Jana), or some statistics; some programming, ideally in Python.
- Curiosity about what a measurement can and cannot tell, interest in experiments at large neutron facilities, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

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